

Worksheet

1. Find the Maclaurin series for $f(x) = 3x^2 \cos(5x)$.
2. Find the Maclaurin series for $\int 3x^2 \cos(5x) dx$.
3. Find the Maclaurin series for the derivative of $3x^2 \cos(5x)$.
4. Find the first 4 terms of the Taylor series for $f(x) = x \ln x$ centered about $a = 2$.
5. If $f(x) = \sum_{k=0}^{\infty} \frac{5k}{k!} x^k$, what is $f''(0)$?
6. Indicate if the following are always true or may be false.
 - (a) ----- If $\sum_{k=0}^{\infty} a_k x^k$ converges when $x = 4$, then it must converge when $x = 3$.
 - (b) ----- If $\lim_{k \rightarrow \infty} a_k = 0$, then $\sum_{k=0}^{\infty} a_k$ converges.